EXPERIMENT 14

Exploring Pro-Code Extension of a Zoho Creator Application Using Zoho Catalyst / Custom Functions

1. Aim

To explore how a Zoho Creator application can be extended using custom functions and Zoho Catalyst to implement advanced business logic and integrate external services.

2. Software Requirements

- Zoho Creator
- Zoho Catalyst (for Catalyst-based extensions)
- Web browser
- Parking Slot Booking System application
- Deluge scripting language

3. Introduction

Low-code platforms simplify application development through visual form builders, reports, and workflow automation. However, complex requirements may need additional programming capabilities.

Pro-code extension combines low-code development with traditional programming. In Zoho Creator, custom functions can implement specialized business logic using Deluge. Zoho Catalyst can provide external backend services and application components when the use case requires capabilities beyond the Creator application.

4. Theory

4.1 Custom Functions

Custom functions are reusable blocks of code that implement business logic. In Zoho Creator, Deluge functions can process records, perform calculations, validate data, and automate application tasks.

4.2 Zoho Catalyst

Zoho Catalyst is a cloud development platform that supports building backend services and applications using code. Depending on the selected services and configuration, it can be used to implement APIs, background processing, and integrations.

4.3 Integration Between Creator and Catalyst

A Creator application can communicate with an external service through an API. The external service processes the request and returns a response. Authentication,

authorization, input validation, error handling, and secure credential management are important parts of this integration.

5. Use Case: Parking Slot Booking System

The existing Parking Slot Booking System manages parking spaces and booking records. A more advanced version could include:

- Automatic booking validation.
- Prevention of duplicate slot bookings.
- Calculation of parking charges.
- Booking confirmation notifications.
- Integration with external payment services.
- Advanced analytics and reporting.

Custom functions can implement suitable business rules inside Creator. Catalyst can be explored for requirements that benefit from a separately developed backend service.

6. Procedure

Step 1: Open the Existing Application

1. Log in to Zoho Creator.
2. Open the Parking Slot Booking System.
3. Navigate to the application builder.

Step 2: Explore Custom Functions

1. Open the application workflow or automation settings.
2. Locate the Functions or Deluge scripting section, if available.
3. Examine how a custom function can be created and invoked from a workflow.
4. Identify a suitable use case, such as validating a booking before confirmation.

Step 3: Design a Booking Validation Function

The following pseudocode illustrates the intended logic:

```
FUNCTION ValidateBooking(slot, date, time):
  IF slot is not available:
    RETURN "Booking unavailable"
  ELSE:
```

CHECK for conflicting bookings

IF a conflict exists:

RETURN "Slot already booked"

ELSE:

RETURN "Booking can proceed"

This is illustrative pseudocode, not executable Deluge code. A working implementation must use the actual form link names, field types, and record-query syntax in the application.

Step 4: Explore Zoho Catalyst

1. Open the Zoho Catalyst developer portal.
2. Examine the available project and backend service options.
3. Identify how a server-side function or API endpoint could validate a parking booking.
4. Review how the service could receive requests and return responses.

Step 5: Plan the Integration

1. Define the data that Creator would send to the external service.
2. Specify the expected response, such as booking approval or rejection.
3. Plan secure authentication and input validation.
4. Consider network failures, invalid requests, and error handling.

Step 6: Evaluate the Extension

Compare the custom-function approach with an external Catalyst service in terms of complexity, maintainability, integration needs, and deployment requirements.

7. Comparison

Feature	Creator Custom Functions	Zoho Catalyst
Main purpose	Application-specific business logic	Code-based backend services and applications
Programming	Deluge	Supported programming languages and runtimes

Integration	Closely connected to Creator workflows	Communicates through APIs and configured integrations
Best suited for	Validation, calculations, and routine automation	External services and more specialized backend processing
Deployment	Within the Creator application	As separately deployed Catalyst components

8. Advantages

- Extends low-code applications with programmable logic.
- Supports reusable business rules.
- Enables integration with external services.
- Helps separate complex backend processing from the user interface.
- Provides flexibility when application requirements become more advanced.

9. Result

The pro-code extension options for the Parking Slot Booking System were explored conceptually. Custom functions were identified as a way to implement application-specific business logic, while Zoho Catalyst was identified as an option for external backend services and API-based integration.

10. Conclusion

Pro-code extensions complement low-code development by enabling more specialized logic and integrations. Zoho Creator custom functions are suitable for many application-level workflows, while Zoho Catalyst can support separately developed backend services. The appropriate approach depends on the complexity, security, scalability, and integration requirements of the application.

11. References

1. Zoho Creator: https://www.zoho.com/creator/
2. Zoho Creator Help: https://help.zoho.com/portal/en/kb/creator
3. Zoho Catalyst: https://catalyst.zoho.com/